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Effects of electron beam irradiation on mechanical properties and nanostructural–morphology of montmorillonite added polyvinyl alcohol composite

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ABSTRACT

This paper is aiming to analyze the effects of electron beam irradiation on the mechanical properties and structural–morphology of nano-sized montmorillonite (MMT) added polyvinyl alcohol (PVOH) composite. MMT particles were added to the PVOH matrix at various loading level that ranges from 0.5 to 4.5 phr MMT and electron beam irradiated with dosages ranging from 6 to 36 kGy. The results showed that tensile strength of MMT added PVOH composites at 1.5 and 2.5 phr MMT were observed marginally higher compare to neat PVOH when irradiation dosages increased to 26 kGy. However, when the concentration of MMT exceeded 2.5 phr, the application of irradiation seems to cause adverse effect to the PVOH–MMT composite. Besides, according to the X-ray diffraction analysis, the application of low irradiation dosage ($\leqslant$16 kGy) has significantly enhanced the intercalation effect of MMT particles at high loading (4.5 phr) in PVOH matrix. This also found to be consistent with the scanning electron microscopy observation where the dispersion of MMT particles in PVOH matrix was noted to be more homogeneous than non-irradiated samples. Further increment in irradiation dosage up to 36 kGy has significantly reduced the crystallinity which indicates the higher radiation energy could rupture the crystallite structures in PVOH matrix.
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1. Introduction

Polyvinyl alcohol (PVOH) is a unique hydrophilic polymer which soluble in water [1]. PVOH is widely used in food, adhesives, biomedical scaffolds, and sizing agent for textile industries [2]. PVOH has high degree of crystallinity due to the small sized hydroxyl (–OH) pendant group which able to fit into the lattice structure promotes formation of highly orientated structure. Meanwhile, PVOH can be divided into two categories, namely fully hydrolyzed and partially hydrolyzed PVOH. The fully hydrolyzed PVOH is usually higher crystalline compared to partially hydrolyzed PVOH. The highly crystalline material is usually more desirable in term of stronger mechanical properties as compared to the latter due to the presence of strong hydrogen bonds that holds the chains together [3]. These strong bonds prevent the mobility of the chains within the matrix, causing it to exhibit higher rigidity and strength.

Most of the time, the existing properties of PVOH is insufficient to fulfill the advance application nowadays. In fact, the mechanical properties of PVOH can be further improved with the composite hybrid method by adding of mineral type reinforcing agents. Such reinforcement is very important especially in adhesive and biomedical scaffolding applications. One of the most suitable types of reinforcing agent is using montmorillonite (MMT). Many researches have been reported that the nano-size MMT is an effective and biocompatible reinforcing which shown its reinforcing effect even applied at a very small quantities [4]. This filler has been explored significantly recently years due to its ability to form highly exfoliated composites when added to polymer materials. The end-product, known as polymer/silicate or composite, can form either the exfoliated or intercalated composite. The terms of intercalated and exfoliated are used to describe the two general classes of MMT that can be prepared. The intercalated structures

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are the well ordered multi-layered structures, where the extended polymer chains are embedded into the interlayer space between the individual of MMT layers. The exfoliated structures form when the individual silicate layers are no longer close enough to interact with the adjacent layers' gallery cations. The interlayer spacing can be on the order of the radius of gyration of the polymer. Therefore, the silicate layer may be considered to well disperse in the organic polymer. The silicate layer of an exfoliated structure may not be as ordered as in intercalated structure. The nanostructures of the MMT can be measured and observed by using X-ray Diffraction measurements. The intercalated and exfoliated nanocomposites shows better mechanical and barrier performances than microcomposites due to its large interfacial surface area and small particle distance of MMT [5].

Commonly, exfoliated composite can be easily obtained compared to intercalated composite because it able to achieve a certain extent of stiffness, strength, and barrier properties as compared to the conventionally filled polymers with small quantity of filler [6]. Nevertheless, studies done by several papers have discovered that intercalated and exfoliated coexist on MMT added polymer nanocomposites [6,7]. This is due to the both intercalation and exfoliation are highly depending on the hydrophilicity and surface energy MMT and type of polymer. Previous study by Strawhecker and Manias [7] found that, the addition of Na^+ MMT into PVOH discovered that MMT has improved the mechanical and thermal properties of the neat PVOH up to a limit before exhibiting inferiority due to agglomeration drawback. In this case, it is expected that further enhancement can be done via the introduction of crosslinking networks. Up to date, many papers reported on the research of PVOH-layered silicate [6–8]. Nevertheless, the studies focused on the characterization of PVOH–MMT composite has not been well explored with application series of electron beam irradiated dosage as the secondary enhancement element on PVOH–MMT compound. Therefore, this paper focuses on the effect of electron beam irradiation on the mechanical properties, thermal properties, as well as the crystallographic structure of the PVOH–MMT composites. Electron beam has been chosen over other forms of irradiation due to its high energy and lower reduction time subsequently produces less embrittlement on the irradiated polymeric compound Almaslow et al. [9].

2. Methodology

2.1. Materials

Fully hydrolyzed polyvinyl alcohol Sekisui Selvol™ polyvinyl alcohol 103 with 4% solution viscosity at 20 °C 4.00 cP and hydrolysis 98.40 mole% manufactured by Sekisui Specialty Chemicals America, LLC. was used in this study. The commercial grade of montmorillonite (MMT), Nanomer 1.3P was purchased from Nanocor, Arlington Height. All these materials were used as received.

2.2. Sample preparation

The samples of PVOH–MMT were prepared with MMT at 0.5, 1.5, 2.5, 3.5 and 4.5 phr. Initially, PVOH was dissolved in distilled water using a water bath at 97 ± 2 °C for 30 min until all PVOH particles have dissolved for one hour. A motor driven stirrer with appropriate speed (300 rpm) was used to stir the mixtures evenly. The stirrer was stopped upon reaching homogenous mixture. The MMT was added into the solution and the stirring was resumed until it completely dissolved. The dissolved solution was the cast onto trays of approximately 1 mm thickness. The cast samples were dried in an oven at 60 °C for two days. The dried cast samples were peeled off from the cast trays and packed in polyethylene bag to keep out from moisture. On the other hand, a portion of cast sample were electron beam irradiated at irradiation voltage of 15 kV, current of 1 mA, and energy of 1 MeV under the EPS-3000 electron beam irradiation machine. The samples were electron beam irradiated at various doses of 6, 16, 26 and 36 kGy with the rate of 2 kGy/pass.

2.3. Samples characterization

2.3.1. Mechanical properties

The cast samples were cut into a rectangular shape of 65.5 mm × 6.0 mm in accordance with ASTM D1882. The cut samples were tested by using Instron tensile machine at crosshead speed of 50 mm/s. The tensile strength, elongation at break and Young modulus were taken as an average of the five specimens.

2.3.2. X-ray Diffractometer (XRD)

The dispersion and crystallinity of MMT in the PVOH matrix were tested by using the XRD-6000 Shimadzu X-ray Diffractometer at 40 kV and 40 mA with the scanning rate of 1°/min. The samples were scanned over a range of 2θ = 0–40°. The d-spacing and inter-chain separation of MMT and MMT added PVOH were calculated with the following formula (Bragg's formula):

$$d = \frac{\lambda}{2\sin\theta} \tag{1}$$

$$R = \frac{5}{8}\left(\frac{\lambda}{\sin\theta}\right) \tag{2}$$

where d = d-spacing, Å; λ = wavelength, nm; θ = Bragg angle, radian; R = inter-chain separation, Å.

The crystallite size of the samples was calculated by using the following formula (Scherrer formula):

$$L = \frac{k\lambda}{b\cos\theta} \tag{3}$$

where L = crystallite size, Å; k = Scherrer constant; b = FWHM, radian.

2.3.3. Scanning Electron Microscopy (SEM)

The surface morphologies of fractured samples were investigated by using a SEM machine of Hitachi S3400N with a voltage of 15 kV. The deformed surface of the samples were observed under the microscopy at magnification of 1000×, 3000× and 10,000×.

2.3.4. Transmission Electron Microscopy (TEM) Test

TEM test was conducted to observe the dispersing and intercalation of MMT particles in polymer matrix. A transmission electron microscope (TEM) with the acceleration voltage of 100 kV was used to study the morphologies of the nano-particles of MMT and the dispersion of MMT in PVOH matrix and its crosslinked constituents. The specimens of samples used for TEM test were in ultrathin form. The ultrathin specimens were sectioned by using a cryogenic ultra-microtome.

3. Results and discussion

3.1. Mechanical properties

Fig. 1 shows the tensile properties of the PVOH–MMT at varying irradiation dosages. Generally, it can be found that when MMT was added at low level <1.5 phr, there is an increment of tensile strength of the composite. This observation is in agreement with the previous observation of various researchers [10] where the intercalation of MMT have occurred that promotes the effective

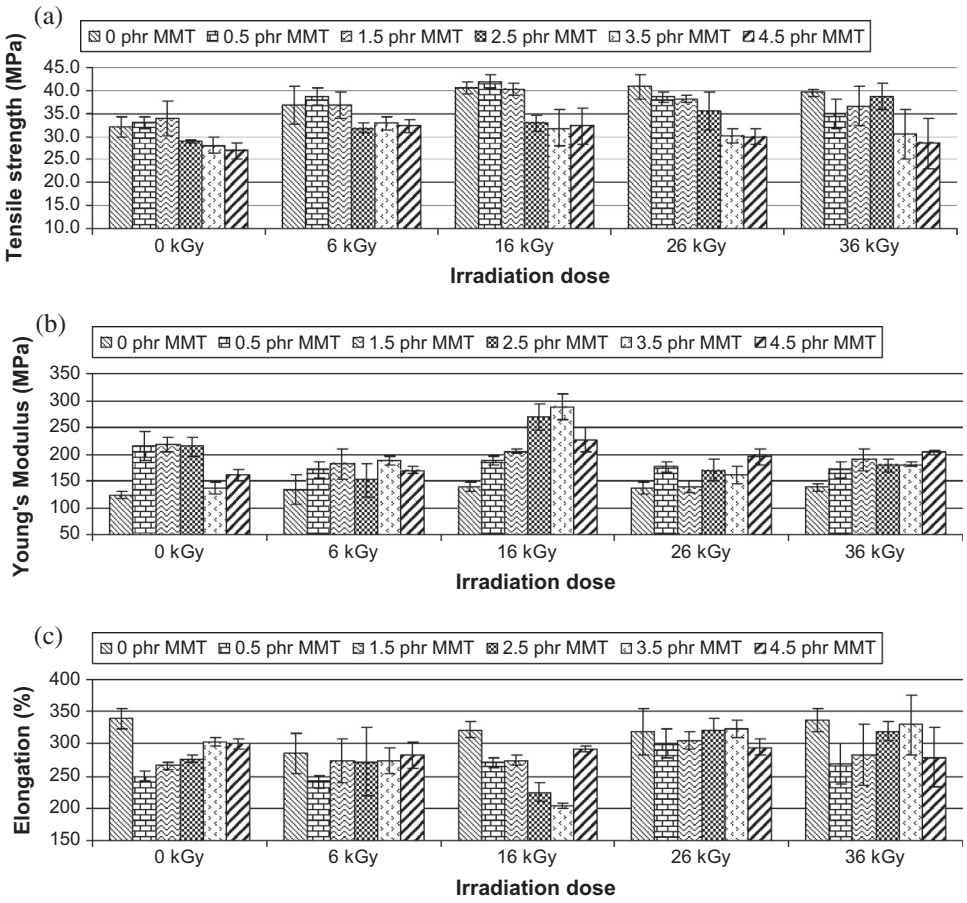


Fig. 1. (a) Tensile strength, (b) Young's modulus and (c) elongation for electron beam irradiated PVOH–MMT compound.

transfer of extension loading to the strong and well ordered multilayered structures of silicates. This is especially important in PVOH matrix which consists vast amount of hydroxyl (–OH) functional group to interact effectively with the inherent hydrophilic characteristic of MMT. The hydrophilic surface of MMT can form hydrogen bonding to induce strong binding forces with the polymer matrix. Nevertheless, when the concentration of MMT has reached the level of 2.5 phr above, the poor dispersion of MMT caused inferior effect to the polymer matrix. In other words, the inverse relationship is due to the non-homogeneous distribution of MMT agglomerates that caused stress concentration on the PVOH matrix when subjected to extension [11]. The phase separation destroyed the continuity of PVOH matrix and further reduced the intercalation effect of PVOH matrix into MMT interlayer galleries which weakened the mechanical performance of the composite [12].

On the other hand, it was found that the application of irradiation can further improve the tensile strength of the neat PVOH and PVOH–MMT. The electron beam irradiation induced the formation of crosslinking network in the polymer matrix which resisted the chains movement when drawing. Initially, the improvement of tensile strength for neat PVOH was significant when the irradiation dosage increase to 16 kGy, further increment up to 36 kGy exhibited slightly reduction of tensile strength. The improvement of tensile strength at lower irradiation dosage might be due to the structural rearrangement among the entangled chains towards more ordered form with the action of crosslinking. Whereas, the inferiority of tensile strength of composite when irradiated at >26 kGy was mainly due to the rupture of PVOH structure as attacked by excessive free radicals upon irradiation at high dosage. Such loss of tensile strength was mentioned by Ahmad et al. [13] that chain scission and cross-linking coexisted especially when irradiation was applied at high dosage. Besides that, the tensile strength of MMT added PVOH composites at 1.5 and 2.5 phr MMT were observed marginally higher compare to neat PVOH when irradiation dosages increased to 26 kGy. The crosslinking that was formed in irradiated PVOH matrix resisted the pulling force which improved the overall tensile strength of the composite as compared to non-irradiated composite [14]. However, when the concentration of MMT exceeded 2.5 phr, the application of irradiation seems to cause adverse effect to the PVOH–MMT composite. This is due to the strong interlayer galleries of MMT distributed all over the polymer matrix has suppressed the free movements of electron beam irradiation free radicals to attack the polymer chains for formation of crosslinking network. This phenomenon is also obviously shown by referring to the results of Young's modulus where the rigidity of PVOH–MMT for both 3.5 and 4.5 phr MMT at high level irradiation dosage of 26 and 36 kGy samples are slightly lower than 0 and 6 kGy irradiated PVOH–MMT composite containing 1.5 and 2.5 phr MMT. The Young's modulus results also showed that the highest rigidity PVOH–MMT can be achieved at the 2.5–3.5 phr MMT and 16 kGy irradiation dosage. At this stage, the intercalation and exfoliation of MMT can take effect while allowing the free movements of radicals to induce formation of crosslinking effectively subsequently enhanced the structure.

Meanwhile, analysis on the elongation showed that neat PVOH has the highest value compared to the others. This is because neat PVOH has non-restricted entangled polymer chains which can slide freely when subjected to extension. The introduction of crosslinking network resulted by electron beam irradiation tended to

restrict chains sliding capability. Nevertheless, it was also observed that the elongation was higher when PVOH–MMT subjected to high irradiation dosage. This can be explained when PVOH–MMT composites were exposed to high irradiation dosages, the chains scissioning process would occur which produced low molecular weight compound. Such shorts chain compound would act as the chains slider to provide ease of movement of the polymer matrix under restricted compact structure. In addition, the subsequent high loading level of MMT which leads to agglomeration would reduce the reinforcing effect of MMT. The decrement in elongation might be caused by agglomeration where MMT particles were unable to distribute evenly which caused clusters of MMT to form. The clusters hinder the smooth flow of PVOH as it elongates thus reduce the elongation of PVOH. On the other hand, it was noted that the elongation increased for PVOH–MMT with higher loading level, i.e. 3.5 and 4.5 phr of MMT compared to neat PVOH. This is due to the lubricating effect of MMT [15] in PVOH matrix allows the chains to slide across each other upon elongation. The hydrophilic characteristic of MMT which provided the inter-chain lubricating effect ([7], Fu et al. [16]) to PVOH chains caused the enhancement of elongation at break in relation to MMT loading level.

3.2. X-ray diffraction analysis

3.2.1. Dispersion of MMT particles

Fig. 2(a) shows the XRD patterns of pristine MMT and PVOH added various loading level of MMT (0.5–4.5 phr) from $2\theta = 0°$ to $2\theta = 3°$. It was found that the pristine MMT exhibits two peaks (shown as area peak A and peak B) on XRD curve at $2\theta = 0.36°$ (peak A) and $2\theta = 1.25°$ (peak B). By referring to Fig. 2(a), the peak A of all MMT added PVOH composites shifted to lower 2θ and nearly disappeared from XRD curves. Besides, such shifting was also observed at peak B for all MMT added PVOH composites as compared to peak B of pristine MMT. This indicates that the PVOH matrix was effectively intercalated into interlayer galleries of MMT particles [17]. Furthermore, the *d*-spacing values (for peak B) of all MMT added PVOH composites (as shown in Table 1) were slightly increased when compared to the peak B of pristine MMT. The increment in *d*-spacing of all MMT added PVOH composites were in agreement with the shifting of peak B. This is mainly due to the intercalation of PVOH matrix into the interlayer galleries could increase the distance between interlayers of MMT particles. The increasing of d-spacing values also further proved the occurrence of effective intercalation of PVOH matrix into interlayer

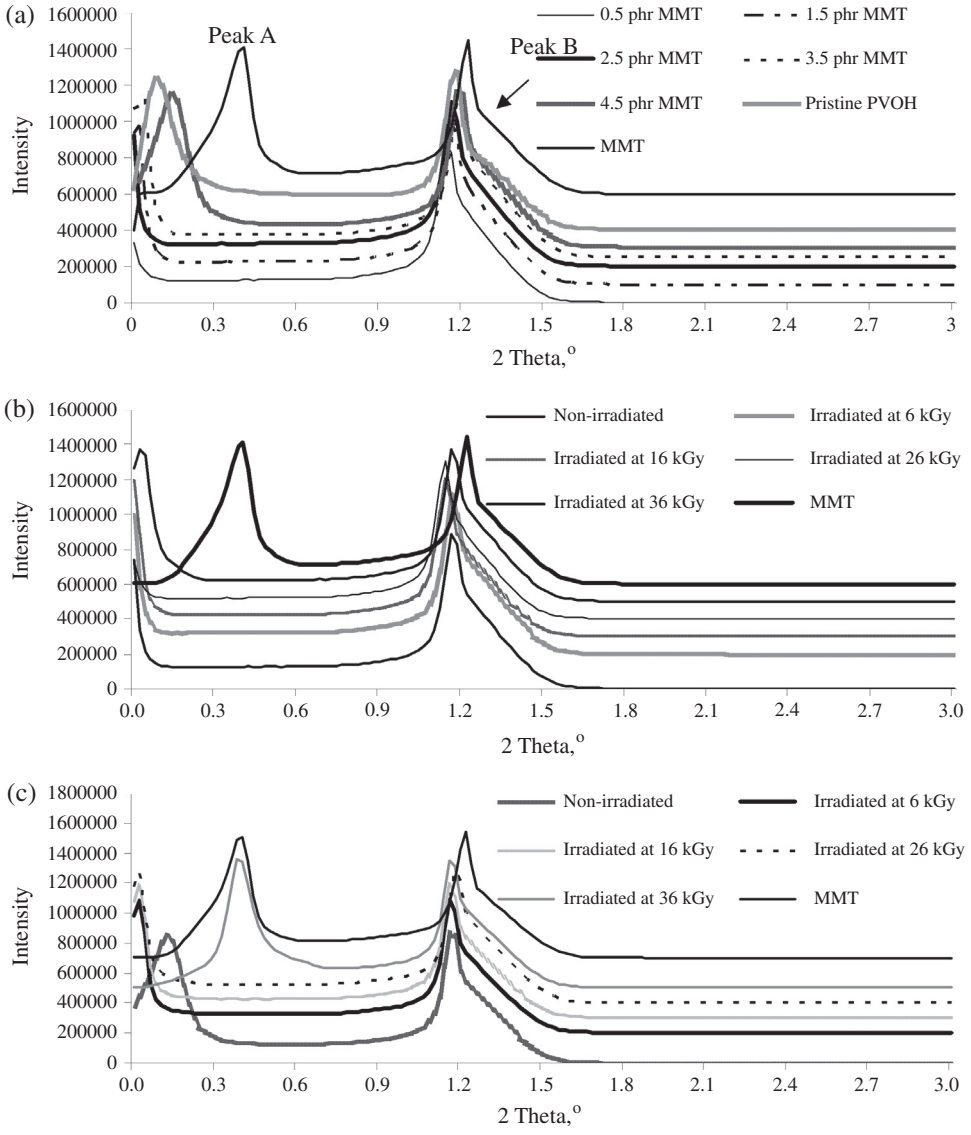


Fig. 2. XRD curves ($0° \leqslant 2\theta \leqslant 3°$) for (a) non-irradiated PVOH added with various loading level of MMT, (b) 0.5 phr MMT added PVOH and (c) 4.5 phr MMT added PVOH composites.

Table 1
Inter-chain separation and *d*-spacing of pristine MMT and MMT added PVOH composites at various irradiation dosages.

Loading level of MMT, phr	Irradiation dosage, kGy	2θ, °	*d*-spacing, Å	Δd, Å	Interchain separation, Å
MMT	–	1.2458	70.9	–	88.6
0.5	0	1.1938	74.0	3.1	92.5
	6	1.1928	74.1	3.2	92.6
	16	1.1763	75.1	4.2	93.4
	26	1.1718	75.4	4.5	94.2
	36	1.2044	73.3	2.4	91.7
1.5	0	1.1965	73.8	2.9	92.3
	6	1.1976	73.8	2.9	92.2
	16	1.1818	74.8	3.9	93.4
	26	1.1742	75.2	4.3	94.0
	36	1.2079	73.3	2.4	91.7
2.5	0	1.2131	72.8	1.9	91.0
	6	1.1982	73.7	2.8	92.2
	16	1.1823	74.6	3.7	93.3
	26	1.2138	72.8	1.9	91.0
	36	1.2166	72.6	1.7	90.8
3.5	0	1.2166	72.6	1.7	90.8
	6	1.1993	73.7	2.8	92.1
	16	1.1795	74.9	4.0	93.6
	26	1.2165	72.6	1.7	90.8
	36	1.2123	72.9	2.0	91.1
4.5	0	1.2216	72.3	1.4	90.4
	6	1.2070	73.2	2.3	91.5
	16	1.1817	74.9	4.0	93.4
	26	1.2182	72.5	1.6	90.7
	36	1.2160	72.6	1.7	90.8

galleries of MMT particles [18]. Meanwhile, the occurrence of effective intercalation of MMT particles in PVOH matrix can also be clearly observed in Fig. 3(a). The distance between the inter-layer galleries of MMT particles has expanded with intercalation of PVOH matrix. Nevertheless, the increasing of MMT loading level in PVOH matrix (from 0.5 to 4.5 phr) has slightly reduced the *d*-spacing value (peak B). This also indicates that the increasing of MMT particles in PVOH matrix could slightly reduce the effective intercalation of MMT particles in PVOH matrix. This might be because of the MMT particles are tended to agglomerate together in PVOH matrix at higher loading level. The agglomeration of MMT particles could reduce the number of effective intercalated MMT particles within PVOH matrix [19]. On the other hand, the inter-chain separation (for peak B) of all MMT added PVOH composites also observed to be slightly reduced with increasing of MMT loading level as shown in Table 1. This also shows that the separation distance between MMT molecules chains in PVOH matrix had been slightly increased. This is an indication that the addition of MMT at various loading level to the PVOH matrix was well distributed.

The effect of electron beam irradiation on 0.5 phr MMT added PVOH composite is shown in Fig. 2(b). The peak B on XRD curves of all irradiated (<36 kGy) 0.5 phr MMT added PVOH composite was observed to slightly shift to lower 2θ when compared to the non-irradiated composite. Besides, the increasing of irradiation dosage (0–26 kGy) has marginally increased the *d*-spacing values (peak B) of 0.5 phr MMT added PVOH composites from 74.0 to 75.4 Å. It was noted that *d*-spacing and inter-chain separation of the composites generally increased with increasing irradiation dose up to 26 kGy. The increment of *d*-spacing and inter-chain separation exhibited that the irradiation cross-linking formed in PVOH matrix could improve the intercalation effect of MMT particles in PVOH matrix. This further indicates that MMT particles were tended to intercalate in PVOH matrix when exposes to low irradiation dose and caused better dispersion pattern of MMT particles in PVOH matrix. Upon application of high irradiation dose of 36 kGy, it was noted that there was a small shift in peak B from 2θ = 1.1938° to 1.2044°. The shifting of peak B to higher 2θ value at high irradiation dosage (36 kGy) signifies a reduction in *d*-spacing as shown in Table 1. The decrement of *d*-spacing and inter-chain separation of 0.5 at higher irradiation doses ($\geqslant$36 kGy) showed that the higher degree of cross-linking networks formed in PVOH matrix do not further improve the intercalation effect of MMT particles in PVOH matrix.

The effect of electron beam irradiation on 4.5 phr MMT added composites (as shown in Fig. 2(a)) was found to be similar to the trend on 0.5 phr MMT added PVOH composites. The *d*-spacing and inter-chain separation of 4.5 phr added PVOH composites were slightly increased as the irradiation dose increased up to 16 kGy. However, further increment in irradiation dosage from 16 to 36 kGy has slightly decreased the *d*-spacing and inter-chain separation of 4.5 phr MMT added PVOH composites. This indicates lower irradiation crosslinking could effectively improve the intercalation of PVOH matrix into MMT particles and dispersion of MMT particles in PVOH matrix. However, further increasing in irradiation crosslinking do not show significant enhancement in intercalation effect of MMT particles in PVOH matrix.

3.2.2. Crystallinity

The effect of various MMT loading level on the crystallinity of PVOH composites is as shown in Fig. 4 and Table 2. According to Table 2, the crystallinity of PVOH composites was slightly decreased from 10.8% to 9.0% when 0.5 phr MMT was added into PVOH matrix. This indicates that the addition of low MMT amount into PVOH matrix could slightly disturb the crystal structural arrangement of PVOH matrix. This is due to the addition of 0.5 phr MMT had significantly shifted the peak A to lower 2θ and nearly disappeared from XRD curve as shown in Fig. 2(a). This also indicates the addition of MMT particles could interrupt and rupture the orientation of crystallite structure (peak A) in polymer matrix of PVOH composites [20]. Besides, a broad peak C was observed to appear on XRD curves of both pristine PVOH and all non-irradiated MMT added PVOH composites as shown in Fig. 4. This might be due to the non-homogeneous distribution of MMT

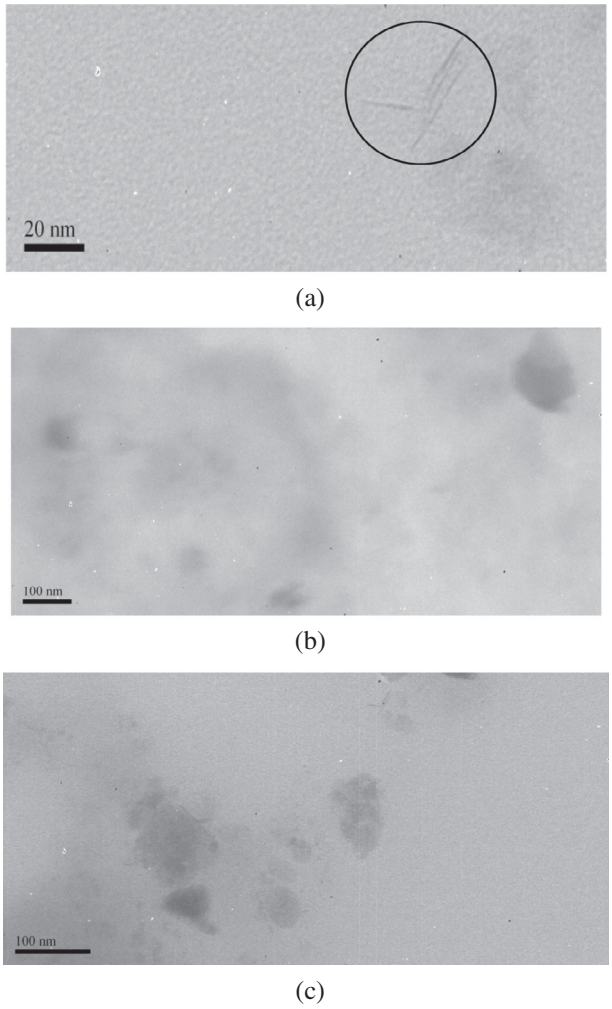


Fig. 3. TEM morphologies observation of (a) the intercalation effect of MMT particles in PVOH matrix, (b) 1.5 phr MMT added PVOH composites under magnification of 30,000 times and (c) 3.5 phr MMT added PVOH composites under magnification of 30,000 times.

which prevents the MMT particles from penetrating into the PVOH matrix. By referring to Fig. 4, pristine PVOH showed a peak C at $2\theta = 19.58°$ whereas 0.5 phr MMT added PVOH composite showed a broader peak C at $2\theta = 19.64°$. The broadening effect on peak C of 0.5 phr MMT added PVOH composites was observed to be more

Table 2
Crystallinity and crystallite size of peak C for pristine PVOH and MMT added PVOH composites at various irradiation doses.

Loading level of MMT, phr	Crystallinity, %				
	0 kGy	6 kGy	16 kGy	26 kGy	36 kGy
0	10.8	10.1	20.1	18.1	10.1
0.5	9.0	11.3	10.4	12.5	18.9
1.5	10.0	15.9	10.0	15.7	16.7
2.5	9.6	15.9	9.7	11.2	9.4
3.5	9.4	9.7	10.7	12.6	10.5
4.5	11.0	16.9	12.2	12.2	6.9
	Crystallite size				
0	43	41	71	40	38
0.5	40	40	41	42	43
1.5	39	85	41	33	41
2.5	43	66	43	43	37
3.5	43	38	39	39	40
4.5	39	78.	40	41	38

significant than pristine PVOH as shown in Fig. 4. The peak broadening effect of peak C exhibited that this crystalline structure in PVOH matrix has been slightly ruptured when 0.5 phr MMT added into PVOH matrix [20]. This peak broadening on peak C as shown in Fig. 4 can be further proved with the crystalline size of peak C for pristine PVOH and 0.5 phr MMT added PVOH composite. The crystalline size (peak C) of pristine PVOH was decreased from 42 to 40 Å when 0.5 phr MMT was added into PVOH matrix. This further shows that the crystallite structure (peak C) in PVOH has slightly re-structured with addition of 0.5 phr of MMT particles. Eventually, this contributed to the lower crystallinity of non-irradiated 0.5 phr MMT added PVOH composites. Besides, the appearance of new small peak G (small crystalline structure) in PVOH matrix did not provide significant increment in crystallinity of 0.5 phr MMT added PVOH composites. When the MMT loading level in PVOH matrix increased up to 4.5 phr, the crystallinity increased up to 11% as shown in Table 2. This is mainly attributed to the poor dispersion of MMT particles in PVOH matrix which the MMT particles tended to agglomerate together when incorporated into the PVOH matrix as discussed in earlier. By referring to Fig. 2(a), the peak A for 4.5 phr MMT added PVOH composites were observed to be appeared at higher $2\theta = 0.13°$ in compared to pristine PVOH ($2\theta = 0.10°$). This also indicated that the agglomerated MMT particles did not disrupt and rupture the crystalline structural arrangement of peak A in PVOH matrix. Furthermore, the peak A of 4.5 phr MMT added PVOH composite was observed to be sharper than peak A of pristine PVOH. This also shows that the expansion in crystallite size for peak A (Fig. 2(a)) in the polymer matrix of

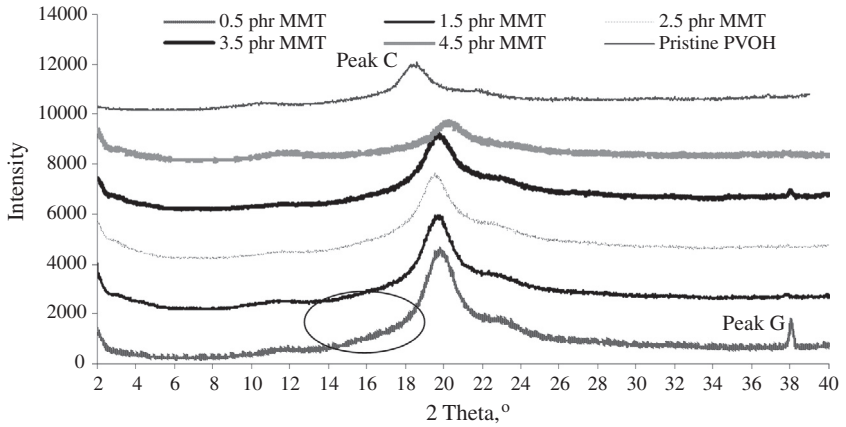


Fig. 4. X-ray diffraction patterns of all non-irradiated pristine PVOH and PVOH composites added with increasing loading level of MMT from $2\theta = 2–40°$.

4.5 phr MMT added PVOH composites. Thus, the 4.5 phr MMT added PVOH composites exhibited the highest crystallinity in comparing to the pristine PVOH and all MMT added PVOH composites even the crystallite size of peak C for 4.5 phr MMT added PVOH composite was decreased to 37 Å (as shown in Table 2).

The effect of electron beam irradiation on the crystallinity of pristine PVOH and all MMT added PVOH composites is shown in Table 2. The crystallinity of pristine PVOH showed a marginal decrement from 10.8% to 10.1% when it was irradiated at the dose of 6 kGy. This is due to the higher broadening effect on peak width of peak C as illustrated in Fig. 5(a), which indicates the breakdown in the crystallite structure of peak C. This observation also found to be consistent with the obtained crystallite size of peak C for 6 kGy irradiated-pristine PVOH (as shown in Table 2). The crystallite size of peak C for pristine PVOH was slightly reduced from 42 to 41 Å when subjected to irradiation dosage of 6 kGy. Thus, the application of low irradiation dosage ($\leqslant$6 kGy) on pristine PVOH did not provide significant effect in crystallinity. By referring to Table 2, the increasing of irradiation dosage from 6 to 16 kGy has exhibited a great increment in the crystallinity of pristine PVOH (from 10.1% to 20.1%). This is consistent with the observation on XRD pattern curve of 16 kGy irradiated pristine PVOH as shown in Fig. 4(a). According to Fig. 4(a), three new peaks, which are peak D, peak E and peak F were observed to appear at $2\theta = 16.12°$, 13.28° and 25.41°, respectively when pristine PVOH was irradiated up to 16 kGy. The appearance of new peaks on XRD pattern curve of 16 kGy irradiated pristine PVOH indicates the formation of three new crystallite structures in pristine PVOH matrix. This indicates that the application of electron beam irradiation dose up to 16 kGy on PVOH matrix able to transform the existing chains arrangement in PVOH matrix into highly ordered crystalline structures [20]. By referring to Fig. 4(a), the peak D (huge in intensity), peak E and peak F were observed to be sharp in peak shape. This also indicates the higher crystallite size of each crystallite structure formed which could significantly contribute to the increment of crystallinity of 16 kGy irradiated pristine PVOH. On the other hand, the peak C of 16 kGy irradiated pristine PVOH as shown in Fig. 4(a) was also observed to be sharper in shape (less broadening effect).

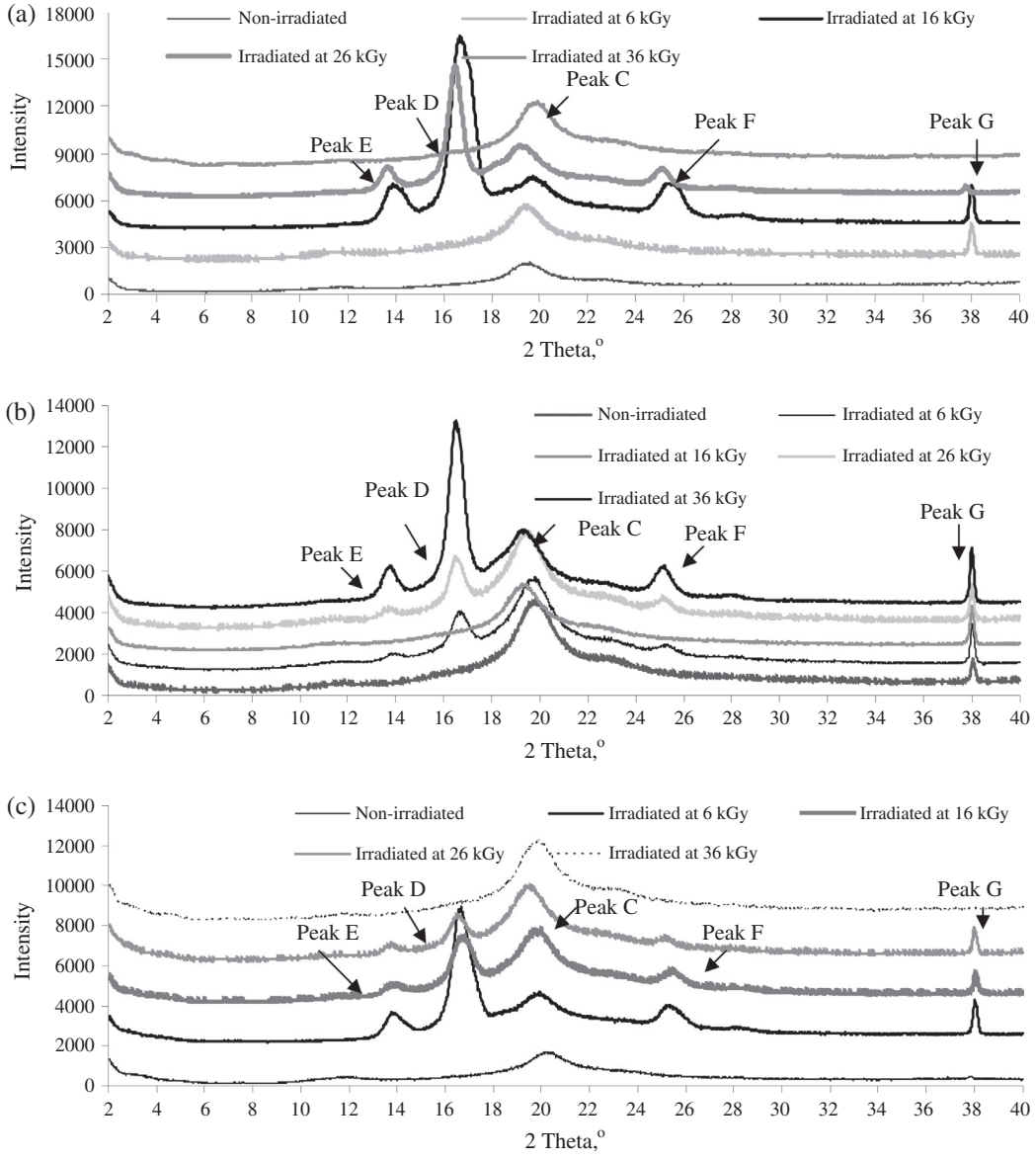


Fig. 5. X-ray diffraction patterns from to $2\theta = 2$–40° of (a) pristine PVOH, (b) 0.5 phr MMT added PVOH composites and (c) 4.5 phr MMT added PVOH composites under increasing of irradiation dosages from 0 kGy to 36 kGy.

This also indicates that further increment of irradiation dosage up to 16 kGy has greatly expanded the crystallite size of peak C from 41 to 71 Å. Thus, the crystallinity of 16 kGy irradiated pristine PVOH was highly increased. However, the increasing of irradiation dosage from 16 to 26 kGy has gradually decreased the crystallinity of pristine PVOH from 20.1% to 18.1%. The highly ordered crystalline structures in pristine PVOH were started to breakdown with further increment in irradiation dose of 26 kGy. This can be observed from XRD pattern curve which the peak intensity of peak D, peak E, peak F and peak G of 26 kGy irradiated pristine PVOH was significantly reduced. The broadening effect on all peaks (C–G) of 26 kGy irradiated pristine PVOH also can be observed in Fig. 4(a). The reduction in peak intensity and the peak broadening effect had led to reduce the crystallinity of 26 kGy irradiated pristine PVOH. Upon to the irradiation dose up to 36 kGy, the crystallinity of pristine PVOH was severely decreased from 18.1% to 10.1%. The severe reduction in crystallinity of polymer matrix is mainly due to the rupture of highly ordered crystalline structures in polymer matrix. This can be further proved by observing the XRD patterns of 36 kGy irradiated pristine PVOH, where the peaks of D, E, F and G were disappeared from XRD patterns. This also indicates that the higher irradiation dosage up to 36 kGy could rupture the highly ordered crystalline structures in PVOH matrix by occurrence of chain scissioning. Thus, the crystallinity of pristine PVOH was severely decreased when subjected to higher irradiation dosage.

The crystallinity of 0.5 phr MMT added PVOH composites shows a gradually increment when subjected to increasing doses of electron beam irradiation as shown in Table 2. XDR patterns of 0.5 phr MMT added PVOH composite at various irradiation doses is as shown in Fig. 4(b). By referring to Fig. 4(b), the non-irradiated MMT added PVOH composite only showed the appearance of a broad peak at $2\theta = 19.82°$ (peak C) and a narrow sharp peak at $2\theta = 38°$ (peak G). As irradiation dose increase to 6 kGy, it was noted that a new peak appeared at $2\theta = 16.46°$ (peak D) and that the intensity of peak G at $2\theta = 38°$ was observed to be increased. The appearance of new peak D indicating the formation of new crystallite structure in the matrix of 0.5 phr MMT added PVOH composite. Thus, this has significantly increased the crystallinity

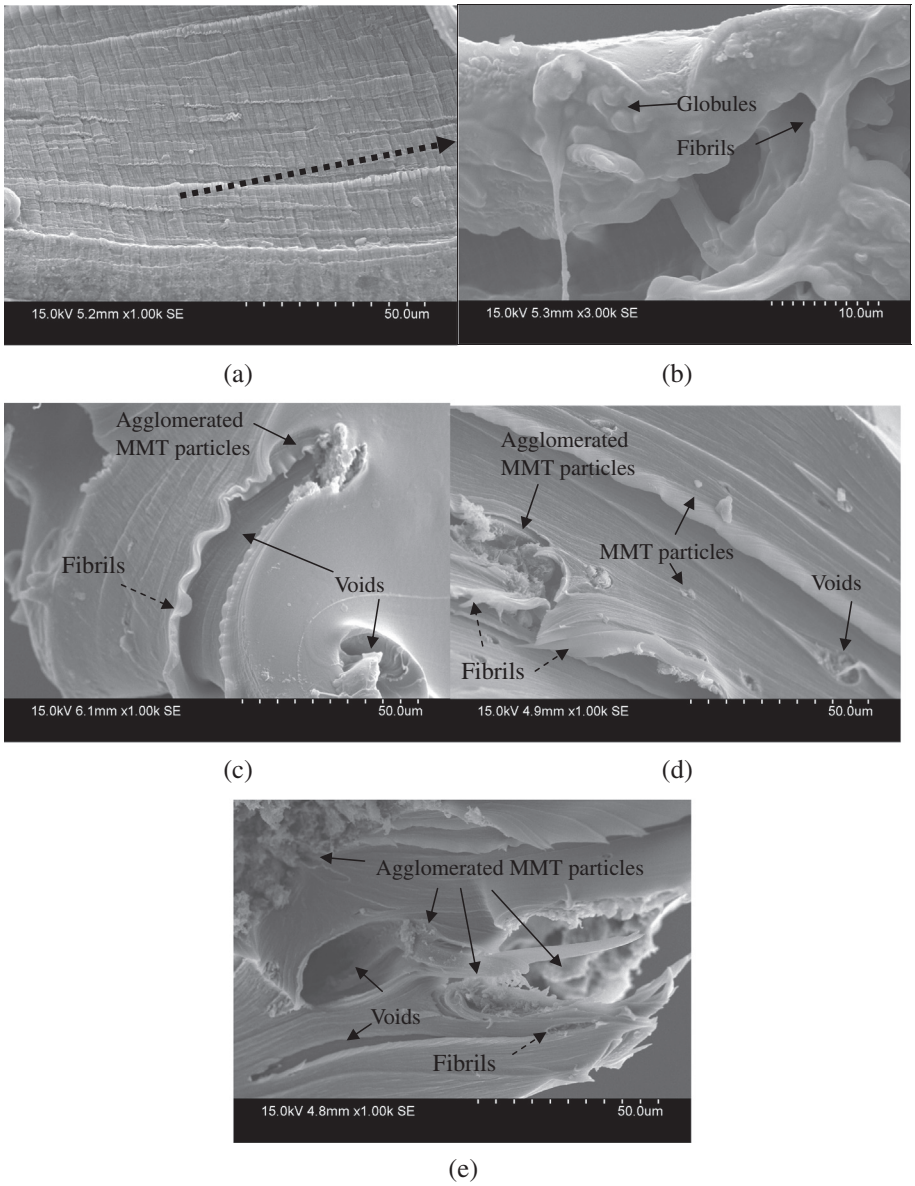


Fig. 6. SEM micrographs of fractured surface for non-irradiated (a) pristine PVOH under magnification of 1000 times, (b) pristine PVOH under magnification of 3000 times and PVOH composites added with (c) 0.5 phr MMT, (d) 1.5 phr MMT and (e) 4.5 phr MMT under magnification of 1000 times.

of 0.5 MMT added PVOH irradiated to 6 kGy. Upon higher irradiation dose of 36 kGy, another two new peaks were appeared at $2\theta = 13.74°$ (peak E) and 25.18° (peak F). Intensity of peaks D and G were observed to enhance sharply as shown in Fig. 4(b). The appearance of new peaks was due to the structural rearrangement of the PVOH chains [21] which might be due to the formation of cross-linking could promote the formation of highly ordered crystal structures in other areas of the base polymer. Similar observations were also reported by Bhat et al. [21] where the crystallinity of new peaks formed upon irradiation increases with increasing irradiation doses.

At higher loading level of MMT (4.5 phr), the crystallinity of MMT added PVOH composites was greatly increased from 11.0% to 16.9% when irradiated to 6 kGy. This result is in agreement with the XRD patterns as portrayed in Fig. 4(c) where the appearance of four new peaks was observed. This indicates that increase in irradiation dose to 6 kGy had significantly increased the crystallinity of 4.5 phr MMT added PVOH composite. It was also noted that the changes of the XRD patterns of 6 kGy irradiated 4.5 phr MMT added sample were significant with the appearance of four new peaks at $2\theta = 20.01°$ (peak C), 16.32° (peak D), 13.82° (peak E), 25.71° (peak F) and 38.32° (peak G). This might be due to the formation of cross-linking networks in MMT added PVOH matrix which have caused the structural rearrangement of PVOH chains in the amorphous region of PVOH matrix to form highly ordered crystallite structures [11]. Furthermore, the crystallite size of peak C for 4.5 phr MMT added PVOH matrix also found to increase from 37 to 79 Å. This also showed that the application of irradiation dose of 6 kGy could re-orientate the chain arrangement around crystallite structure of peak C and expand the crystallite size of peak C [20]. However, further increasing the irradiation dose up to 36 kGy has rapidly reduced the crystallinity of 4.5 phr MMT added PVOH composite from 16.9% to 6.9%. This result can be further explained by referring to Fig. 4(c) which all the peaks D to G were found to be disappeared from XRD patterns of 4.5 phr MMT added PVOH composite. This indicates that the higher irradiation dose of 36 kGy could lead to breakdown the highly ordered chain arrangement of crystallite structures. This is due to the occurrence of chain

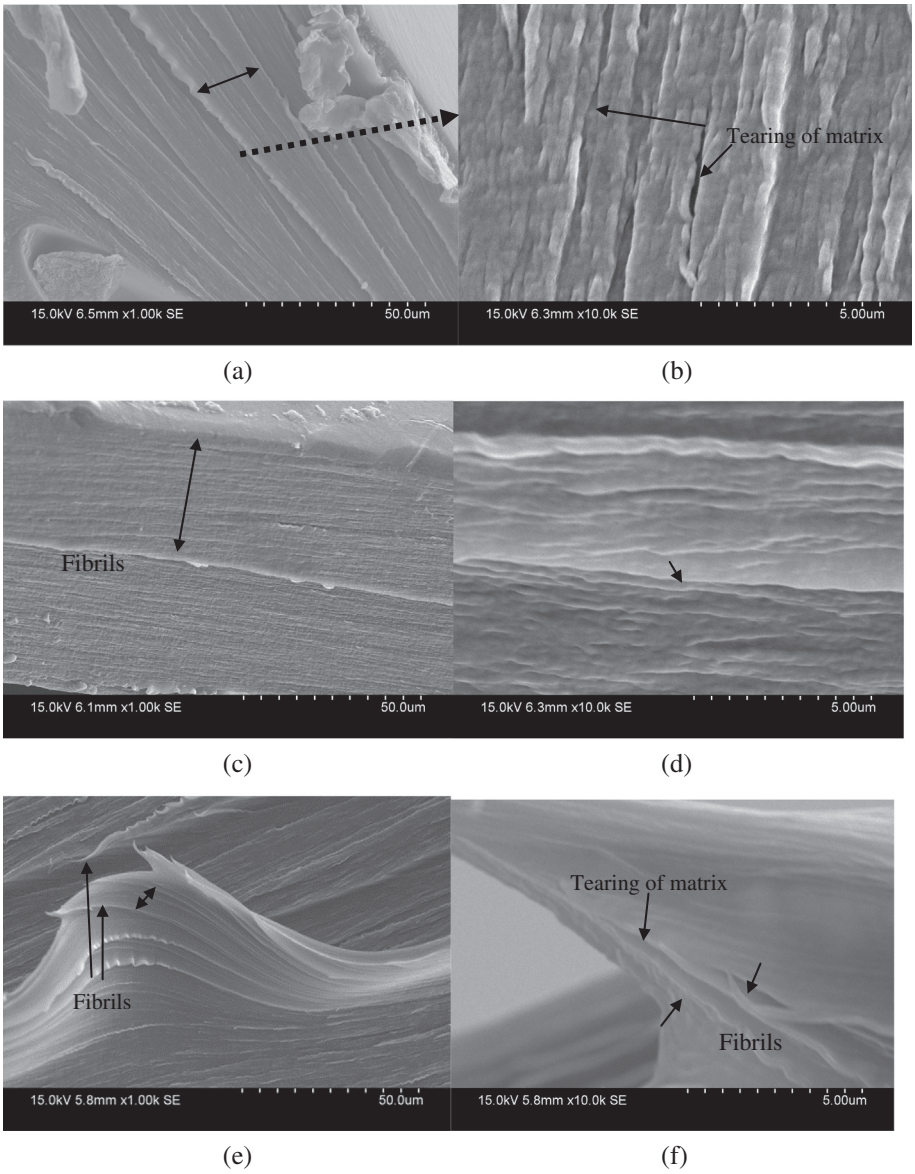


Fig. 7. SEM micrographs of fractured surface for pristine PVOH under irradiation dosages of (a) 6 kGy at magnification of 1000 times, 6 kGy at magnification of 10000 times, (c) 16 kGy at magnification of 1000 times (d) 16 kGy at magnification of 10,000 times, (e) 36 kGy at magnification of 1000 times and (f) 36 kGy at magnification of 10,000 times.

scissioning in 4.5 phr MMT added PVOH matrix where the chain scissioning process could breakdown the crystallite structures [20].

3.3. Surface morphologies observations

Fig. 6 shows the fractured surface morphologies of non-irradiated pristine PVOH and all MMT added PVOH composites. The appearance of globules as observed in Fig. 6(b) was due to the elasticity effect of PVOH when the chains realign themselves after being elongated. Besides, the formation of fibrils in PVOH matrix also can be clearly observed in Fig. 6(b). The formation of fibrils is due to the resistance ability of PVOH matrix from elongating when subjected to straining effect. A similar observation was found for 0.5 phr MMT, 1.5 phr MMT and 4.5 phr MMT added PVOH composites as indicated in Fig. 6(c)–(e). On the other hand, the presence of flakes like groups was also observed in the matrix of all MMT added PVOH composites as depicted in Fig. 6(c), (d) and (e). The presence of flakes in MMT added PVOH matrix was attributed to the non-homogenous distribution of MMT particles in PVOH matrix which tended to agglomerate together to form larger MMT aggregates particles. However, presence of flakes (agglomerated particles) was also observed to be the least in the matrix of 0.5 phr MMT added PVOH when comparing to 1.5 phr and 4.5 phr MMT added PVOH samples. This also indicates that the distribution of MMT particles in lower MMT ($\leqslant$0.5 phr) added PVOH composite was also observed to be more homogenous than in higher MMT ($\geqslant$1.5 phr) added PVOH composites. By referring to Fig. 5, the agglomeration effect of MMT particles was also observed to increase drastically when the MMT loading level in PVOH matrix increased to 1.5 phr and 4.5 phr MMT. Similar observation also can be seen in TEM micrographs as depicted in Fig. 3(b) and (c). By referring to Fig. 3(b) and (c), the agglomerated MMT particles in PVOH matrix were observed to be significantly increased when the MMT loading level increased from 1.5 to 3.5 phr. This also indicates that the increasing of MMT loading level in PVOH matrix could promote the agglomeration effect of MMT

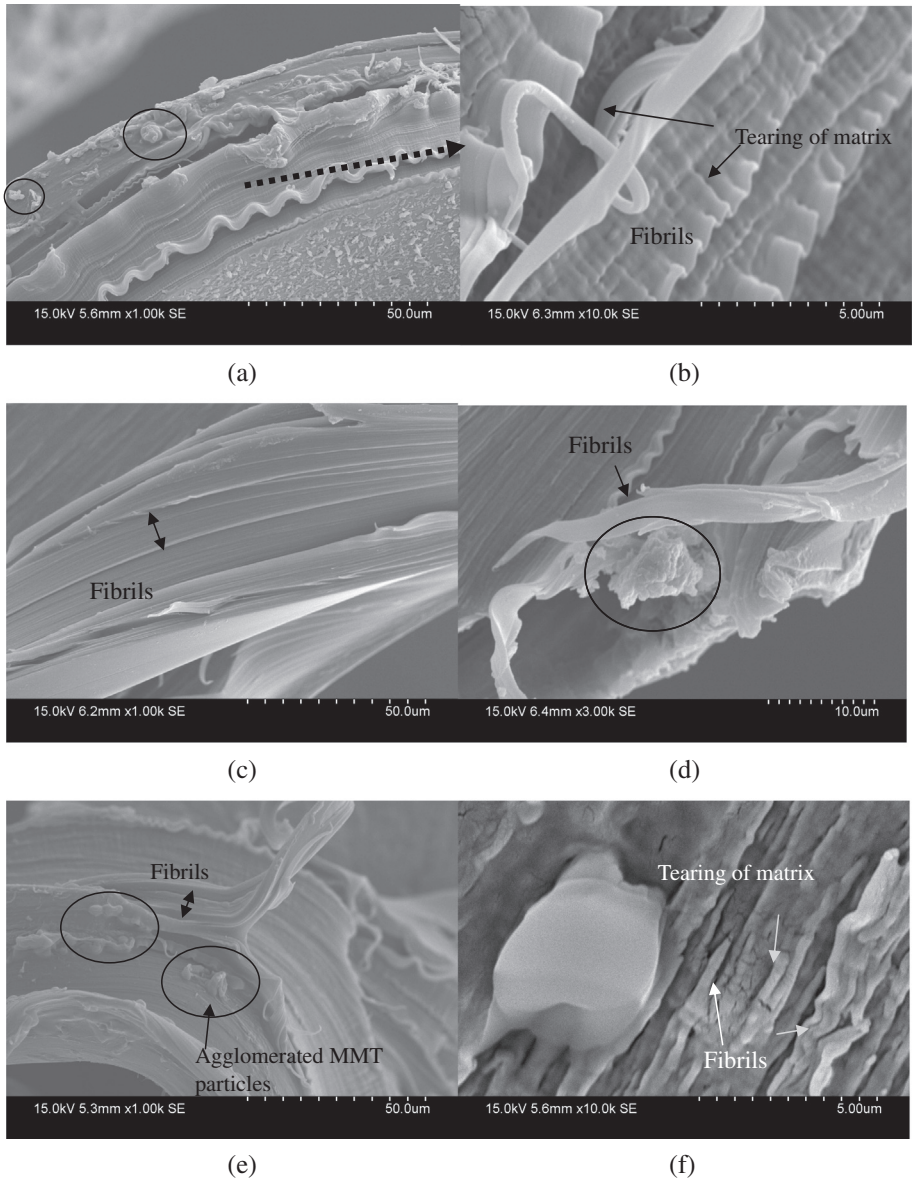


Fig. 8. SEM micrographs of fractured surface for 0.5 phr MMT added PVOH under irradiation dosages of (a) 6 kGy at magnification of 1000 times, 6 kGy at magnification of 10,000 times, (c) 16 kGy at magnification of 1000 times (d) 16 kGy at magnification of 10,000 times, (e) 36 kGy at magnification of 1000 times and (f) 36 kGy at magnification of 10,000 times.

particles in PVOH matrix. Agglomeration effect caused by the non-homogeneous distribution of MMT particles in PVOH matrix also could act as a hindrance to the mobility of slippage in PVOH chains during re-crystallization [11].

Fractured surface morphologies of pristine PVOH at various irradiation doses are as depicted in Figs. 6(a) and (b) and 7. Fibrils were seen to form in the PVOH matrix of for all non-irradiated and irradiated PVOH. Formation of fibrils was due to the stress that was applied to the PVOH during tensile test [19]. As observed in Fig. 7(a) and (b), the fibrils in non-irradiated PVOH are thinner and higher in fibril amount. As the irradiation dosage increased up to 6 kGy, the width of fibrils formed was observed to significantly increase and the fibrils also found to orientate straightly and parallel to each other. Similar observation also found when the irradiation dosage increased from 6 kGy to 16 kGy. The width of fibrils for 16 kGy irradiated PVOH was observed o highly extended as shown in Fig. 7(c). This is due to the formation of radiation induced crosslinking networks in PVOH matrix could resist the tearing effect of PVOH matrix by extending the continuities of macromolecules chains [22]. The reduction of fibril amount at high irradiation dose is attributed to the formation of cross-linking in PVOH chains that could increase the matrix continuities [23]. Upon high irradiation (36 kGy), the fibrils were observed to be thinner and higher in amount than 6 kGy and 16 kGy irradiated pristine PVOH. However, the width of fibrils formed was observed to be thicker than non-irradiated PVOH (Fig. 7(e) and (f)). This might be due to breakdown of the PVOH crosslinked chains by higher irradiation energy which causing the chain scissioning of PVOH matrix.

The fractured surface morphologies of 0.5 phr MMT added PVOH composite under irradiation dosages of 6 kGy, 16 kGy and 36 kGy is depicted in Fig. 8. As discussed in earlier (as shown in Fig. 6(c)), the agglomeration of MMT particles was observed to occur for non-irradiated 0.5 phr MMT added PVOH composite. However, the size of MMT agglomerated particles appeared in matrix of 0.5 phr MMT added PVOH is found to be smaller when

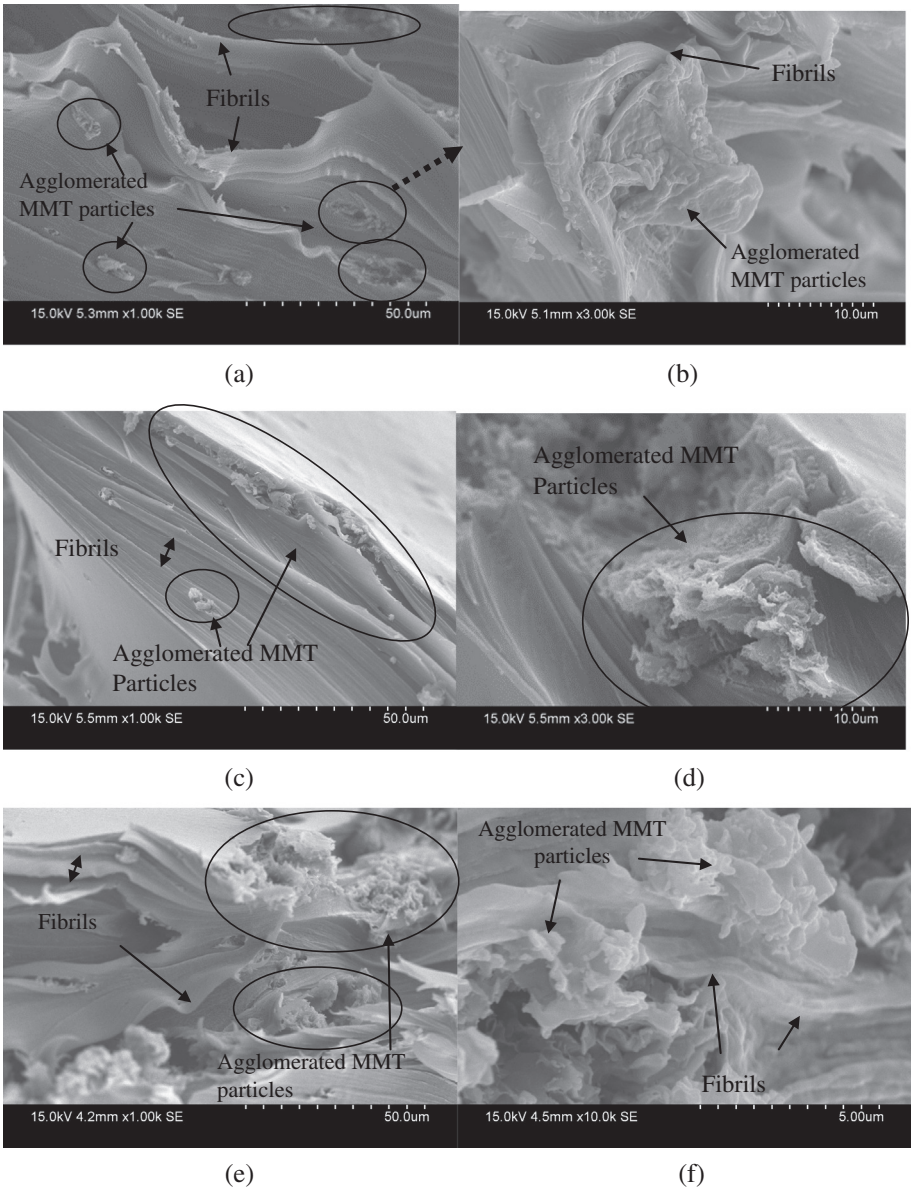


Fig. 9. SEM micrographs of fractured surface for 4.5 phr MMT added PVOH composites under irradiation dosages of (a) 6 kGy at magnification of 1000 times, (b) 6 kGy at magnification of 10,000 times, (c) 16 kGy at magnification of 1000 times (d) 16 kGy at magnification of 3000 times, (e) 36 kGy at magnification of 1000 times and (f) 36 kGy at magnification of 10,000 times.

irradiated to 6 kGy (as illustrated in Fig. 8(a) and (b)). This also indicates that the formation of radiation induced crosslinking networks could reduce the agglomeration effect of MMT particles by enhancing the MMT intercalation and dispersion effect in PVOH matrix. Besides, the fibrils formed also found to thicker in width when compared to non-irradiated 0.5 phr MMT added PVOH. Similar observation also can be seen on the surface of 16 kGy irradiated 0.5 phr MMT added PVOH as depicted in Fig. 8(c) and (d). The occurrence of agglomerated MMT particles was observed disappear from surface matrix of 0.5 phr MMT added PVOH when irradiated up to 16 kGy. The formation of fibrils also observed to less in amount and thicker in width. This further proved that the application of suitable dosage of electron beam irradiation could enhance the intercalation and dispersion effect of MMT in polymer matrix. However, at higher irradiation dosage ($\geqslant$36 kGy), the occurrence of agglomerated MMT particles can be observed in the matrix of 0.5 phr MMT added PVOH as shown in Fig. 8(e). The size of agglomerated MMT particles was also observed to be smaller than non-irradiated samples. By referring to Fig. 8(e) and (f), the width of fibrils formed was significantly reduced and the tearing effect of matrix also can be clearly observed. This also indicates that the application of higher irradiation dosage ($\geqslant$36 kGy) could cause the occurrence of chain scissioning effect which is also detriment to mechanical properties of PVOH matrix. The observations can be further supported by the results obtained from the mechanical test.

The fractured surface morphologies of 4.5 phr MMT added PVOH composites under various irradiation dosages are illustrated in Fig. 9. Upon low irradiation dose of 6 kGy, fibril diameter was seen to be greatly enhanced as compared to non-irradiated 4.5 phr MMT added PVOH composite. This might be due to the formation of cross-linking network [23] that resulted in the increase in the matrix continuity of PVOH. Besides, the agglomeration effect of MMT particles also observed to greatly reduce due to a significant reduction in the size and amount of agglomerated MMT particles in PVOH matrix when irradiated to 6 kGy. This also indicates the application of low irradiation dose could enhance the dispersion and intercalation of MMT particles in PVOH matrix. However, upon higher irradiation dose of 16 kGy and 36 kGy, the fibrils formed were seen to decrease as compared to 6 kGy irradiated 4.5 phr MMT added PVOH composite. Phase separation of MMT particles (flakes) and PVOH matrix was observed as portrayed in Fig. 9(c) to (f) which indicates MMT agglomeration throughout the PVOH matrix. This also indicates that the application of higher irradiation dosage ($\geqslant$16 kGy) did not provide any enhancement effect on the dispersion and intercalation of MMT particles in PVOH matrix when added with higher MMT loading level.

4. Conclusions

The effect of electron beam irradiation dosage on mechanical properties and nanostructural–morphology of PVOH composites added with increasing of MMT loading level has been investigated and led to following conclusion:

1. It was found that the application of irradiation can improve the tensile strength of the neat PVOH and PVOH–MMT. This is due to the electron beam irradiation has promoted the formation of crosslinking network in the matrix. The improvement of tensile strength for neat PVOH was significant when the irradiation dosage increased to 16 kGy, further increment up to 36 kGy exhibited slightly reduction of tensile strength. Upon the application of low irradiation dosage, the structural rearrangement among the entangled chains might be happened towards more ordered form with the action of crosslinking. Meanwhile, the inferiority of tensile strength of composite when irradiated at >26 kGy was mainly due to the rupture of PVOH structure as attacked by excessive free radicals upon irradiation at high dosage. The tensile strength of MMT added PVOH composites at 1.5 and 2.5 phr MMT were observed marginally higher compare to neat PVOH when irradiation dosages increased to 26 kGy. The crosslinking that was formed in irradiated PVOH matrix resisted the pulling force which improved the overall tensile strength of the composite as compared to non-irradiated composite. However, when the concentration of MMT exceeded 2.5 phr, the application of irradiation seems to cause adverse effect to the PVOH–MMT composite. This is due to the strong interlayer galleries of MMT distributed all over the polymer matrix has suppressed the free movements of electron beam irradiation free radicals to attack the polymer chains for formation of crosslinking network.
2. The addition of low MMT loading level (0.5 phr) has effectively intercalated into PVOH matrix as indicated by the increment in d-spacing and inter-chain separation of PVOH composites. However, the intercalation effect (*d*-spacing and inter-chain separation) of MMT particles in PVOH matrix was significantly reduced when the MMT loading level increased up to 4.5 phr. This is due to higher loading level of MMT particles tended to agglomerate together in PVOH matrix and reduced the total amounts of effective intercalated MMT particles. On the other hand, the *d*-spacing and inter-chain separation of low MMT (0.5 phr) added PVOH have gradually increased with increasing of irradiation dosages. This indicates the formation of radiation induced crosslinking networks in PVOH matrix could significantly promote the intercalation effect of MMT particles in PVOH matrix. At higher MMT loading level (4.5 phr), the application of low irradiation dosage ($\leqslant$16 kGy) has significantly enhanced the intercalation effect of MMT particles in PVOH matrix. This also found to be consistent with the SEM observation where the dispersion of MMT particles in PVOH matrix was observed to be more homogeneous than non-irradiated samples. However, the agglomeration of MMT particles in 4.5 phr MMT added PVOH was observed to significantly increase with further increment of irradiation dose up to 36 kGy.
3. The addition of MMT ($\leqslant$3.5 phr) in PVOH matrix has slightly reduced the crystallinity of PVOH matrix by disrupting the crystallite structure in PVOH matrix. The crystallinity of pristine PVOH has gradually increased from 10.8% to 20.1% when irradiated up to 16 kGy. This is due to the formation of radiation induced crosslinking networks in PVOH matrix could transform the existing chains arrangement into highly ordered crystallite structure as indicated in XRD patterns. However, further increment in irradiation dosage up to 36 kGy has significantly reduced the crystallinity which indicates the higher radiation energy could rupture the crystallite structures in PVOH matrix. At low MMT loading level ($\leqslant$1.5 phr), the crystallinity of MMT added PVOH was gradually increased when irradiation increased up to 36 kGy.

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